

Zomato Order & Restaurant Analysis Using Power BI



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Project Objectives

- Analyze restaurant performance across different cities.
- Identify cities with the highest restaurant density.
- Understand customer ordering patterns.
- Analyze revenue contribution by different areas.
- Identify top-performing restaurants.
- Build an interactive dashboard for business analysis.



Dataset Overview

- **2 Tables:** Zomato_Restaurants & Orders
 - **Restaurant Table:** Restaurant details, location, ratings & pricing
 - **Orders Table:** Order, customer, delivery & transaction details
 - **Primary Key:** restaurant_id
 - **Foreign Key:** restaurant_id in Orders
 - **Relationship: One-to-Many (1:*)**
 - **Model:** Restaurant → Orders
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SQL Operations Performed

- Removed duplicate records to maintain data accuracy.
- Checked and handled NULL values.
- Verified data consistency across columns.
- Converted data into appropriate data types.
- Grouped restaurant records based on cities.
- Calculated total revenue.
- Counted total orders.
- Calculated average order value.
- Created summary tables for Power BI visualization.



Results obtained from SQL Queries

Result Grid | Filter Rows

	city	total_orders
▶	Mumbai	39
	Kolkata	27
	Delhi	26
	Chennai	22
	Bangalore	20

city	average_order_amount
Mumbai	972.311026
Kolkata	1040.235926
Delhi	948.380000
Bangalore	1012.550500
Chennai	1036.736364

Result Grid | Filter Rows:

restaurant_name	total_revenue
Restaurant_339	3590.58
Restaurant_14	3364.37
Restaurant_48	2387.34
Restaurant_149	2356.55
Restaurant_392	2223.12
Restaurant_315	2097.40
Restaurant_65	2006.80
Restaurant_341	1984.39
Restaurant_236	1912.67
Restaurant_224	1907.63
Restaurant_44	1890.63
Restaurant_468	1879.86
Restaurant_217	1852.84

restaurant_name	total_sales
Restaurant_339	3590.58
Restaurant_14	3364.37
Restaurant_48	2387.34
Restaurant_149	2356.55
Restaurant_392	2223.12

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	restaurant_name	city	customer_id	order_date	order_time	total_cost	item_count	payment_method	customer_r
▶ 00550aa0	Restaurant_15	Mumbai	4037152	2024-04-06	17:30:00	392.92	6	Wallet	1.5
0411f03f	Restaurant_339	Kolkata	47145167	2024-04-26	01:30:00	1804.13	9	UPI	4.6
045d62d3	Restaurant_73	Kolkata	32318288	2024-10-08	18:35:00	694.24	1	Net Banking	2.5
09f10059	Restaurant_182	Delhi	93059021	2024-10-22	13:58:00	1378.39	1	Net Banking	4.6
0a1c6046	Restaurant_371	Delhi	84819287	2024-02-03	13:20:00	690.53	1	Wallet	4.2
0b2bb392	Restaurant_452	Bangalore	46870101	2024-12-03	23:44:00	1561.12	1	COD	4.8
0fa29b38	Restaurant_392	Kolkata	33350426	2024-06-17	05:32:00	1754.02	10	Credit Card	2.7
114a1cc2	Restaurant_380	Mumbai	1316524	2024-05-24	00:52:00	1171.71	4	Net Banking	3.5
11b27cd2	Restaurant_35	Delhi	94227291	2024-01-26	23:16:00	201.67	10	UPI	1.6
13419b52	Restaurant_299	Chennai	45534313	2024-05-28	02:55:00	1530.40	5	COD	3.0
148930b8	Restaurant_430	Mumbai	44494412	2024-05-17	23:01:00	179.19	7	Credit Card	1.9
1b245455	Restaurant_413	Kolkata	98598913	2024-12-21	07:11:00	245.14	2	COD	4.2
1c3d6b6a	Restaurant_463	Kolkata	68402548	2024-10-17	05:51:00	517.29	8	Net Banking	4.4

Data Preparation & Modeling

Data Preparation

- **Power Query** used for data extraction & transformation
- **Missing values & errors** validated
- **Duplicates** checked and handled
- **Data types** verified & corrected
- **Columns** cleaned and standardized
- **Data quality** validated before analysis

Data Modeling

- Connected both tables using **restaurant_id**
- Maintained **One-to-Many (1:*) relationship**
- Built an **analysis-ready data model**
- Used the model to create **visualizations & dashboard**

Dashboard Objective

Power BI was used to transform cleaned data into meaningful visual reports. Different charts were created to answer specific business questions.

Visualizations Created

- KPI Cards
- Bar Chart
- Pie Chart
- Line Chart
- Treemap
- Heat Map
- Restaurant-wise Summary Table



KPI Analysis

Total Revenue

₹15.57 M

Insight

The dashboard shows that the business generated approximately **₹15.57M** in total revenue during the selected period.

Average Order Value

₹1.04 K

Insight

The average amount spent by a customer per order is approximately **₹1.04K**. This metric helps understand customer spending behavior.

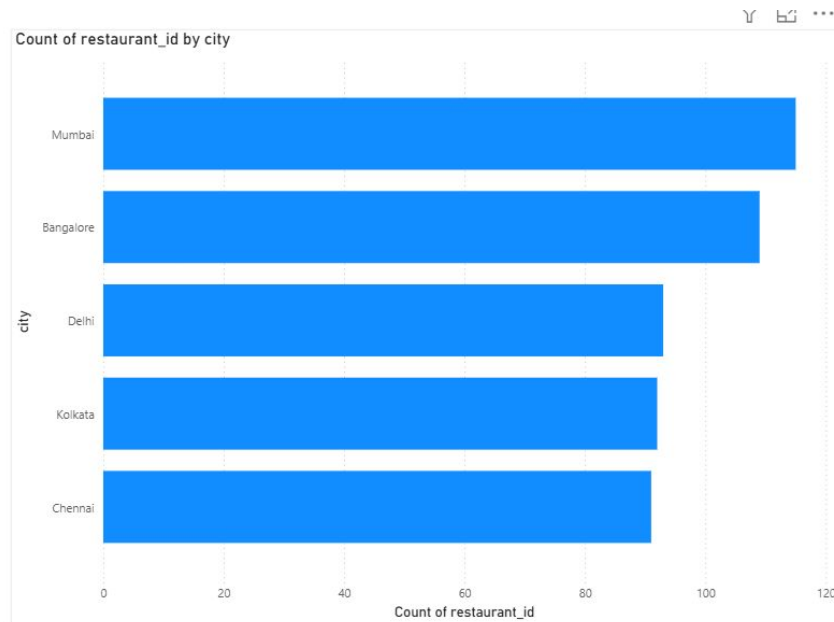
City-wise Analysis

Restaurants Per City (Bar Chart)

Insight

The bar chart compares the number of restaurants available in different cities.

- Mumbai has the highest number of restaurants.
- Chennai has the lowest restaurant count.
- Restaurant availability varies across cities, indicating differences in market presence.



Percentage of Orders by City (Pie Chart)

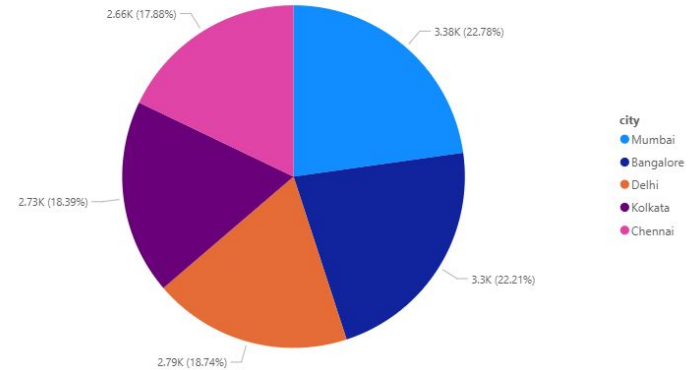
Insight

The pie chart represents the percentage contribution of orders from each city.

- Mumbai contributes the highest share of total orders.
- Bangalore and Delhi also contribute significantly.
- Chennai contributes the lowest percentage of orders.

This visualization helps understand customer demand across cities.

Percentage of Orders from Different Cities
Count of order_id by city



Order Density Heat Map

Insight

The heat map uses color intensity to represent order density.

- Darker colors indicate higher order volumes.
- Mumbai and Bangalore show the highest order density.
- Chennai shows relatively lower order activity.

This visualization quickly identifies areas with high customer demand.

Order Density by City Using Heat Map

city	Count of order_id
Mumbai	3384
Bangalore	3300
Delhi	2785
Kolkata	2732
Chennai	2657

Revenue and Sales Analysis

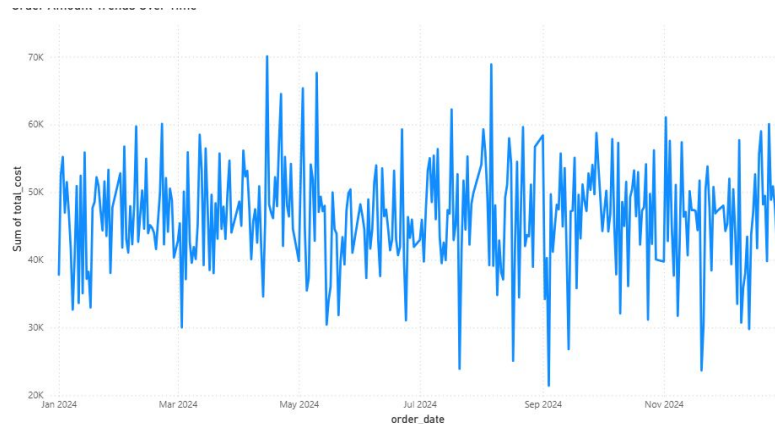
Order Amount Trend (Line Chart)

Insight

The line chart illustrates changes in order amounts over time.

- Sales fluctuate throughout the year.
- Certain months show significant increases in revenue.
- Seasonal demand and promotional activities may influence these fluctuations.

This helps businesses identify peak sales periods.



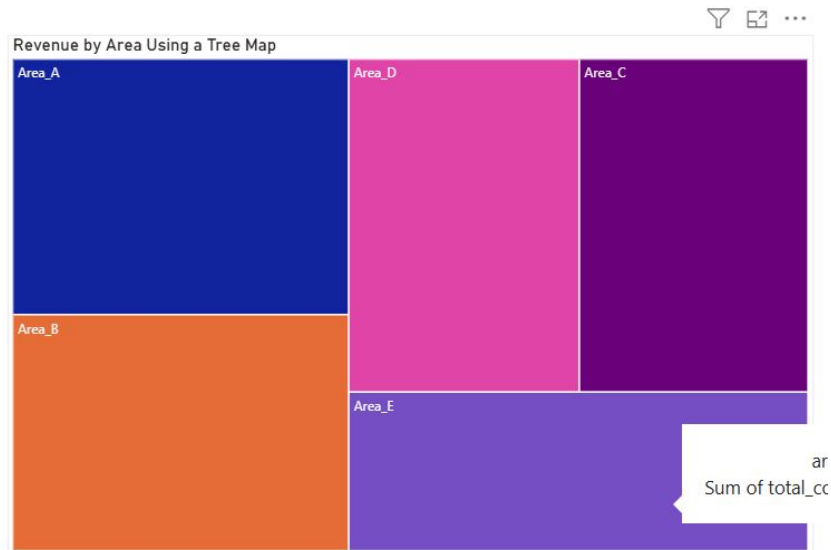
Revenue by Area (Treemap)

Insight

The treemap compares revenue generated by different cities.

- Area A contributes the largest share of revenue.
- Area B, D and C also contribute significantly.
- Area E contributes comparatively less revenue.

The treemap makes it easy to compare Area-wise business performance.



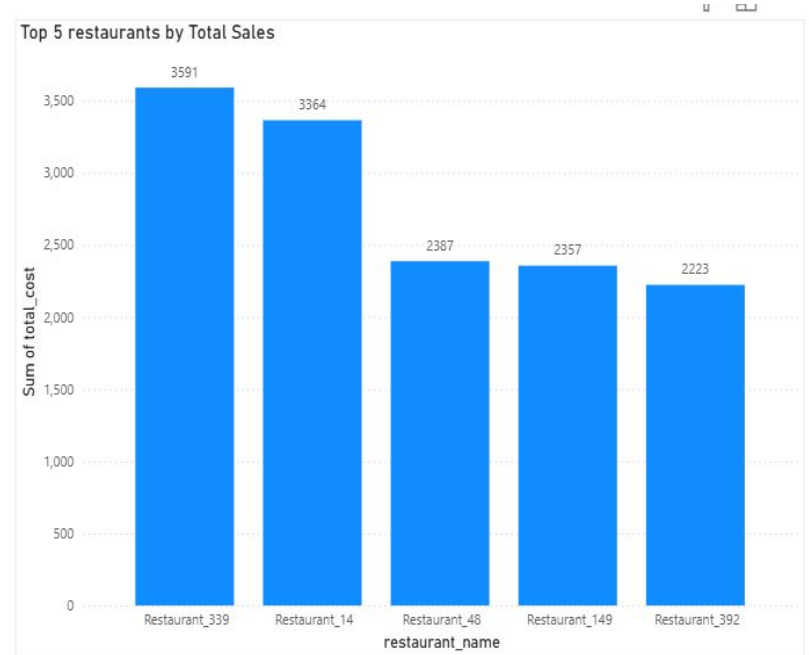
Top 5 Restaurants by Sales

Insight

The bar chart highlights the restaurants generating the highest revenue.

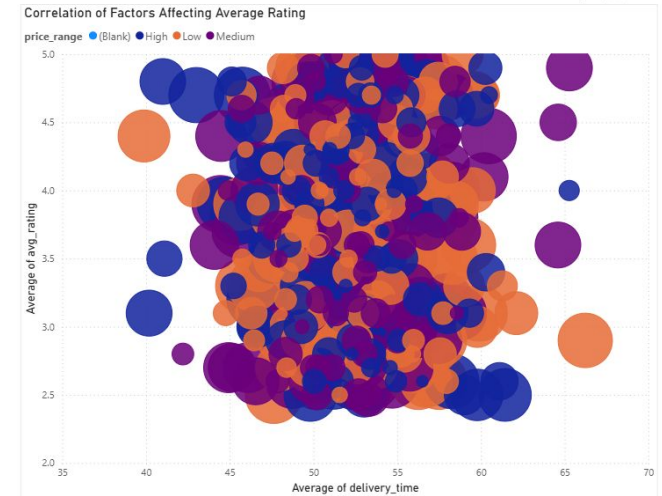
- Restaurant_339 achieved the highest total sales.
- Restaurant_14 secured the second position.
- The remaining restaurants contributed comparatively lower revenue.

This helps identify high-performing restaurants for business planning.



Correlation of Factors Affecting Average Rating

- The relationship between delivery time and restaurant ratings is identified
- A **weak negative correlation** can be observed.
- Restaurants with longer delivery times tend to have slightly lower ratings.



Restaurant-wise Sales Summary

Table Visualization

The table summarizes restaurant performance using multiple business metrics.

It displays:

- Restaurant Name
- Total Sales
- Number of Orders
- Average Order Value

Insight

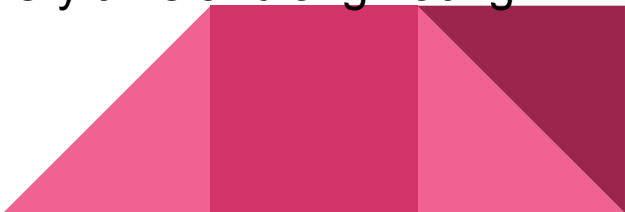
The table provides a detailed comparison of restaurant performance.

It helps businesses:

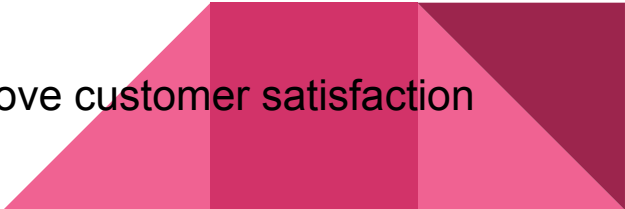
- Identify top-performing restaurants.
- Compare sales across restaurants.
- Monitor order frequency.
- Evaluate average customer spending.

Observation

From this analysis, we observed that:

- Total Revenue reached **₹15.57M**.
 - Average Order Value is approximately **₹1.04K**.
 - Mumbai contributes the highest revenue and order volume.
 - Chennai performs least in all aspects.
 - Restaurant_339 is the highest-performing restaurant.
 - Sales fluctuate over different time periods.
 - Weak negative Correlation is observed between delivery time and avg. rating.
- 

Business Recommendations

- Focus marketing campaigns on **cities with lower order volumes** to increase customer engagement.
 - Analyze the strategies of **top-performing restaurants** and implement similar practices in other locations.
 - Introduce **combo offers and premium meal options** to increase the average order value.
 - Improve inventory and delivery planning by using **order trends and demand patterns**.
 - Expand restaurant partnerships in **high-demand cities** to meet growing customer needs.
 - Continuously monitor business performance through the dashboard to make **faster, data-driven decisions**.
 - Restaurants should focus on reducing delivery time to improve customer satisfaction and potentially increase average ratings.
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Zomato Order & Restaurant Analysis Dashboard

Total Revenue

15.57M

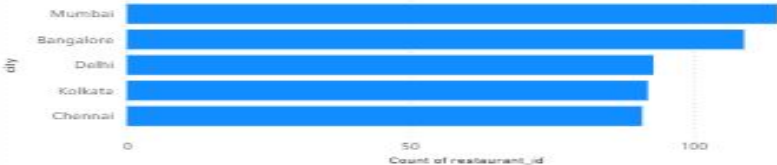
Sum of total_cost

Average Order value

1.04K

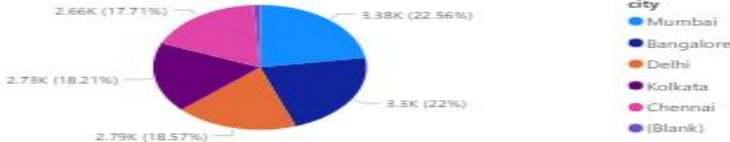
Average of total_cost

Number of restaurant by city



Percentage of Orders from Different Cities

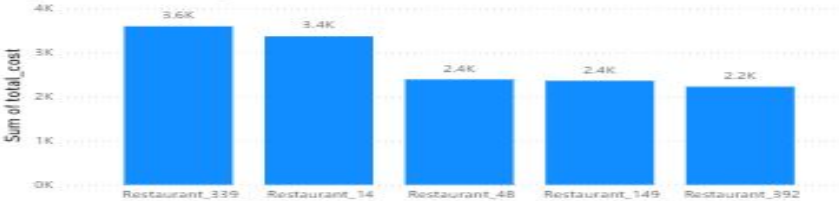
Count of order_id by city



Order Amount Trends Over Time



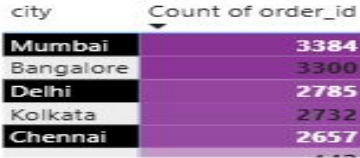
Top 5 restaurants by Total Sales



Revenue by Area Using a Tree Map



Order Density by City Using Heat Map



restaurant_name	Sum of total_cost	Count of order_id	Average of total_cost
Restaurant_0	1,48,303.75	142	1,044.39
Restaurant_1	36,138.83	35	1,032.54
Restaurant_10	33,078.21	35	945.09
Restaurant_100	22,954.14	21	1,093.05
Restaurant_101	22,005.54	25	880.22
Restaurant_101	34,983.35	36	971.76
Restaurant_102	26,155.36	24	1,089.81
Total	1,55,73,886.64	15000	1,038.26

THANK YOU

